torch.autograd.forward_ad.make_dual#

https://pytorch.org/docs/stable/generated/torch.autograd.forward_ad.make_dua

Description:

Associate a tensor value with its tangent to create a “dual tensor” for forward AD gradient computation. The result is a new tensor aliased to tensor with tangent embedded as an attribute as-is if it has the same storage layout or copied otherwise. The tangent attribute can be recovered with unpack_dual(). This function is backward differentiable. Given a function f whose jacobian is J, it allows one to compute the Jacobian-vector product (jvp) between J and a given vector v as follows.

Function Signature

```
torch.autograd.forward_ad.make_dual(tensor, tangent, *,
level=None)[source]#
```

Code Examples

```
>>> with dual_level():
...     inp = make_dual(x, v)
...     out = f(inp)
...     y, jvp = unpack_dual(out)
```

Detailed Documentation

Documentation

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Given a function f whose jacobian is J, it allows one to compute the Jacobian-vector product (jvp) between J and a given vector v as follows.

Please see the forward-mode AD tutorial for detailed steps on how to use this API.